Intro to Database and SQL


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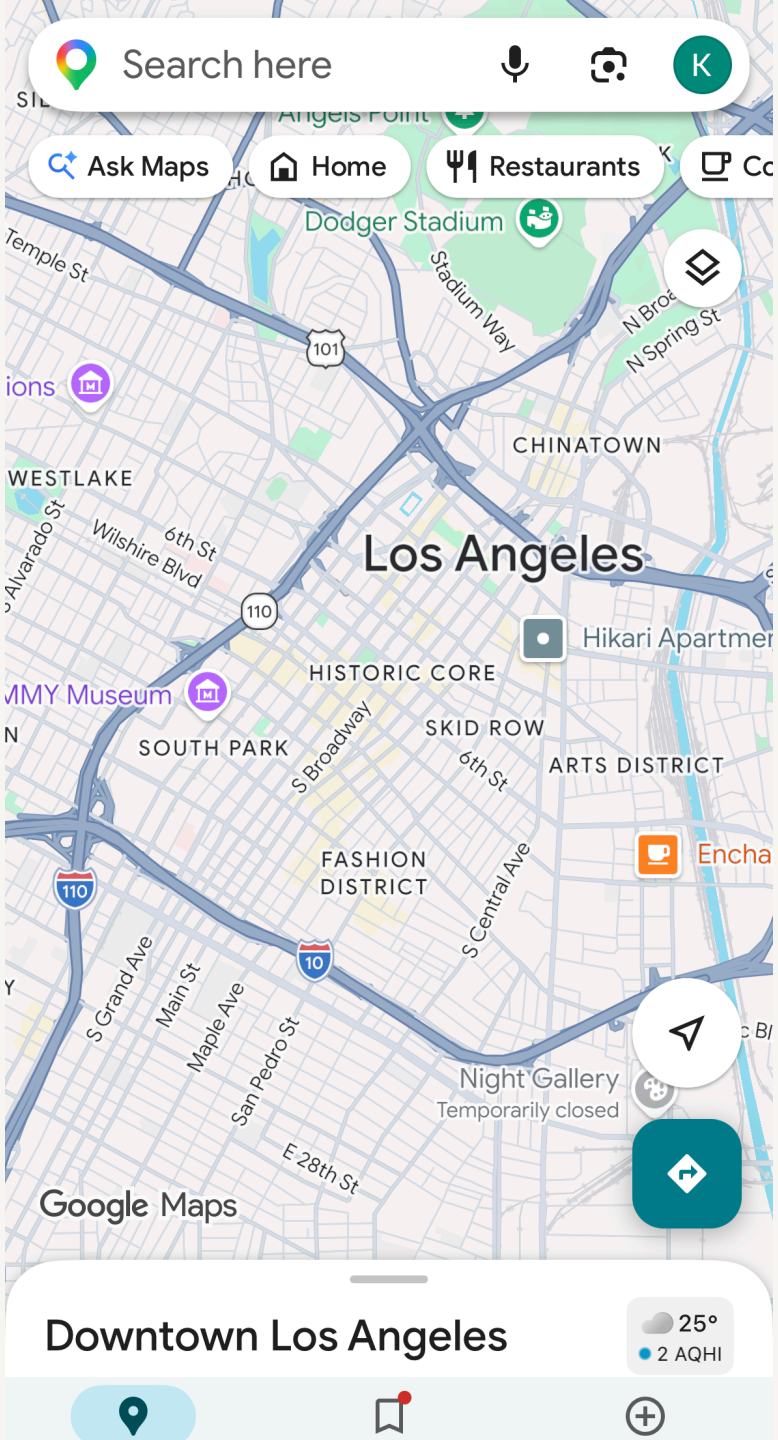
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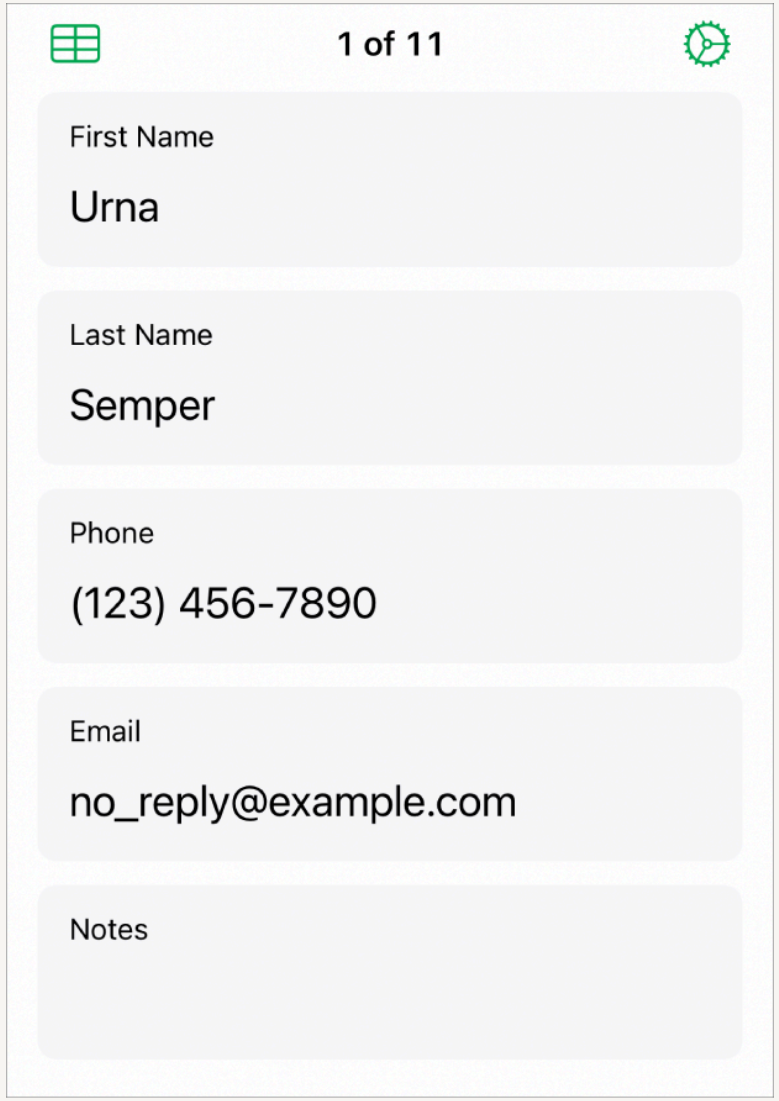
What Is a Database?

> A **structured collection of data** that’s easy to **store, update, and search**.

> Database =
> 🧠 **organized data** + ⚙️ **software that manages it (DBMS)**



Why Do We Need Databases?

> **“Which customer bought Product A on January?”**

Without a database, you’d need to:

1. Open and merge dozens of invoices manually.
2. Check for inconsistent column names and duplicate records.
3. Search through thousands of rows to find the right data.
4. Risk errors whenever someone updates an old file.

Databases to the Rescue

- Store all sales records in **one structured place**.
- Query data instantly using SQL:

```sql
SELECT customer_name
FROM sales
WHERE product = 'A' AND month = 'January';
```

Why Do We Need Databases?

- **Fast answers:**
  Query huge data sets in seconds instead of hours.
- **Keeps data clean and accurate:**
  Built-in rules prevent errors and duplicates.
- **Grows with you (Scalable):**
  Start small on a laptop → scale to millions of users.
- **Works for everyone (Shared access):**
  Multiple people or apps can use the same data safely.

Two Key Concepts to Understand Databases

- **Entity Relationship Diagram (ERD)**
- **Structured Query Language (SQL)**

Entity Relationship Diagram (ERD)

A **blueprint** of a database: how data is organized and connected

Two elements:

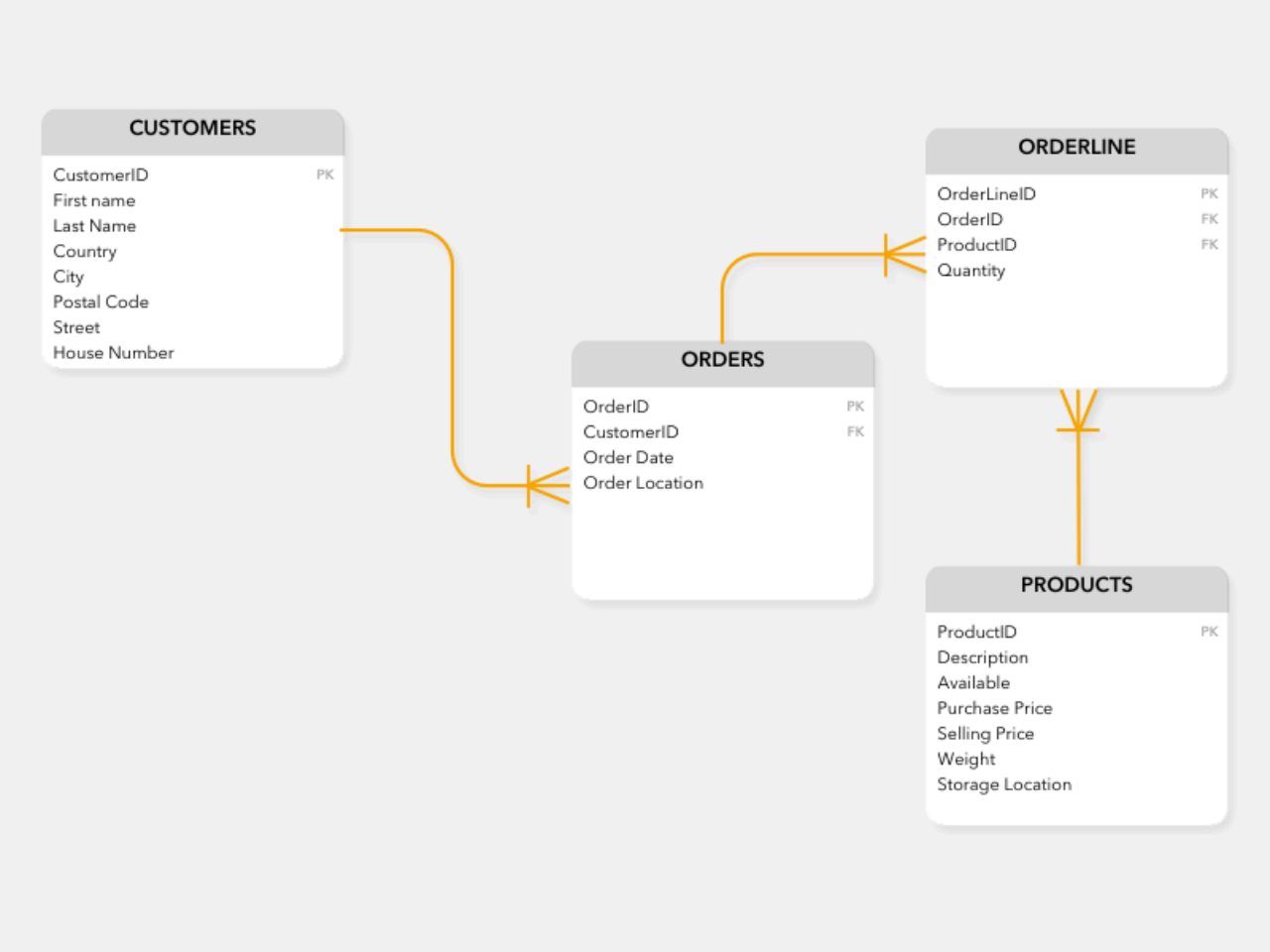
- **Entity (noun):** Something we store data about
  → e.g., `Student`, `Course`, `Enrollment`
- **Relationship (verb):** How entities are linked
  → e.g., `Student` "enrolls in" `Course`

Sample ERD

Entities: Customers, Orders, Products, Orderline

Relationships:

- Customers place Orders
- Orders contain Products via Orderline



Structured Query Language (SQL)

To interact with databases

```
SELECT [ALL/DISTINCT] column_list
FROM table_list
[WHERE conditional expression]
[GROUP BY group_by_column_list]
[HAVING conditional expression]
[ORDER BY order_by_column_list]
```

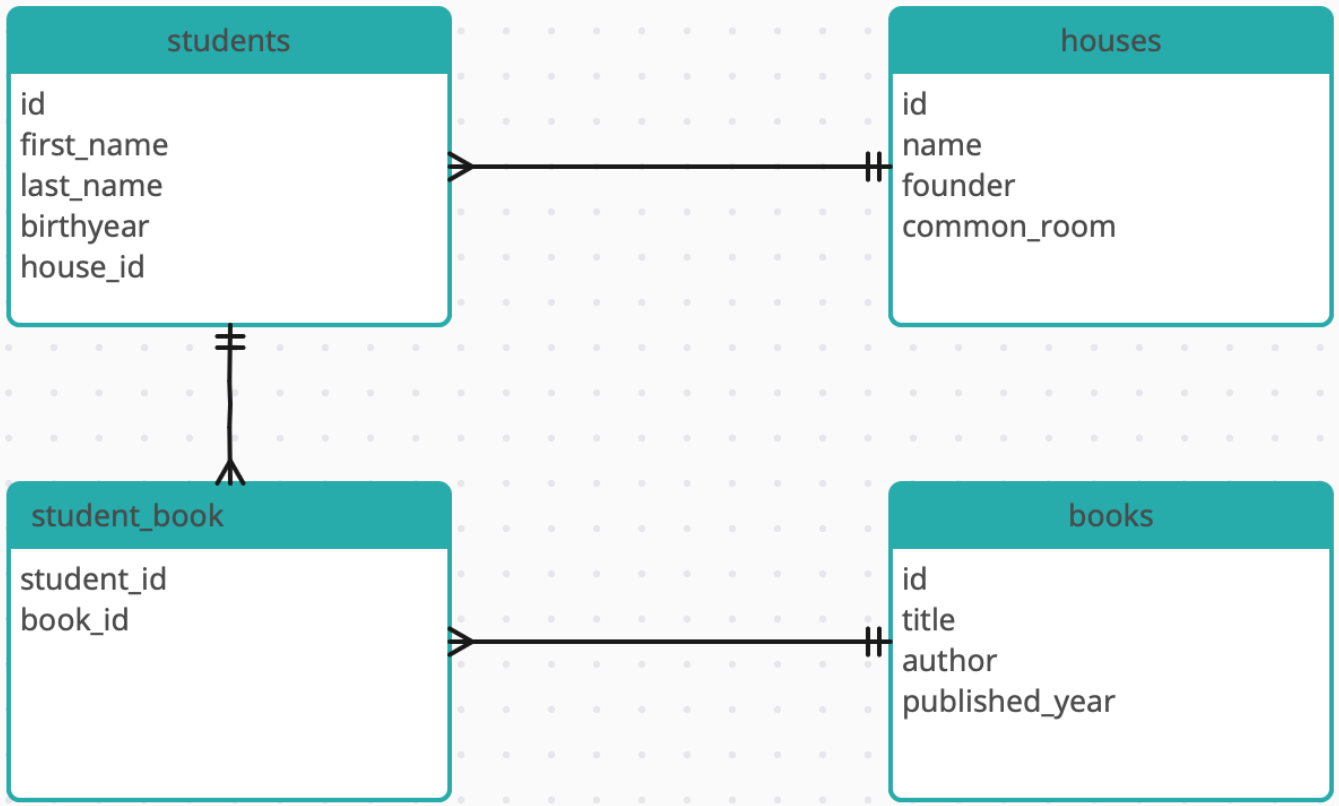

Essentials of Database Management, 1st ed., Pearson.

How to Design a Database

1️⃣ What do you want to store data about? (→Entities)

2️⃣ How are entities connected? (→Relationships)

Harry Potter Database - ERD



`SELECT` columns `FROM` a table

SELECT `column_name` FROM `table_name`

```
-- select first_name column from students table
SELECT first_name FROM students;

-- select first_name and last_name columns from students table
SELECT first_name, last_name FROM students;

-- select all columns from students table
SELECT * FROM students;
```

`*` is a wildcard that means "all columns"

`--` to add comments for your reference. Anything after `--` will be ignored by the database.

- ***dash dash space*** `--`, not ***dash dash*** `--`
- `CMD` + `/` to comment out a line on Mac, `CTRL` + `/` on Windows.

`;` to indicate the end of a SQL statement

Optional if you have only one statement.

`LIMIT` the number of records returned

SELECT `column_name` FROM `table_name` LIMIT `number`

```
SELECT * FROM students LIMIT 10;
```

`SELECT` columns `FROM` a table `WHERE` conditions are true

SELECT `column_name` FROM `table_name` WHERE `condition`

```
-- select first_name and last_name of all students in house_id = 1
SELECT first_name, last_name FROM students WHERE house_id = 1;

-- select first_name and last_name of all students born in 1980
SELECT first_name, last_name FROM students WHERE birthyear = 1980;

-- select first_name and last_name of all students born in or after 1980
SELECT first_name, last_name FROM students WHERE birthyear >= 1980;
```

💻 Write SQL queries to answer the following questions

1. What year was Harry Potter born?
2. List all books that were published after 1999.
3. Who is the founder of Gryffindor?